```
import pandas as pd

df = pd.read_csv('spotify_millsongdata.csv')

df.head()
```

	artist	song	link	text
0	ABBA	Ahe's My Kind Of Girl	/a/abba/ahes+my+kind+of+girl_20598417.html	Look at her face, it's a wonderful face \r\nA...
1	ABBA	Andante, Andante	/a/abba/andante+andante_20002708.html	Take it easy with me, please \r\nTouch me gen...
2	ABBA	As Good As New	/a/abba/as+good+as+new_20003033.html	I'll never know why I had to go \r\nWhy I had...
3	ABBA	Bang	/a/abba/bang_20598415.html	Making somebody happy is a question of give an...
4	ABBA	Bang-A-Boomerang	/a/abba/bang+a+boomerang_20002668.html	Making somebody happy is a question of give an...

```
df.columns
```

```
Index(['artist', 'song', 'link', 'text'], dtype='object')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 57650 entries, 0 to 57649
Data columns (total 4 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   artist  57650 non-null  object
 1   song    57650 non-null  object
 2   link    57650 non-null  object
 3   text    57650 non-null  object
dtypes: object(4)
memory usage: 1.8+ MB
```

```
# Check missing values
df.isnull().sum()
```

	0
artist	0
song	0
link	0
text	0

dtype: int64

```
# Remove missing values
df = df.dropna()

# Check dataset size
df.shape
```

```
(57650, 4)
```

```
!pip install textblob
```

```
Requirement already satisfied: textblob in /usr/local/lib/python3.12/dist-packages (0.19.0)
Requirement already satisfied: nltk>=3.9 in /usr/local/lib/python3.12/dist-packages (from textblob) (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk>=3.9->textblob) (8.3.3)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk>=3.9->textblob) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk>=3.9->textblob) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk>=3.9->textblob) (4.67.3)
```

```
from textblob import TextBlob

# Function to calculate sentiment
def get_sentiment(text):
    analysis = TextBlob(str(text))

    if analysis.sentiment.polarity > 0:
```

```
        return "Positive"
    elif analysis.sentiment.polarity < 0:
        return "Negative"
    else:
        return "Neutral"

# Apply sentiment analysis
df['Sentiment'] = df['text'].apply(get_sentiment)

# View results
df[['song', 'artist', 'Sentiment']].head()
```

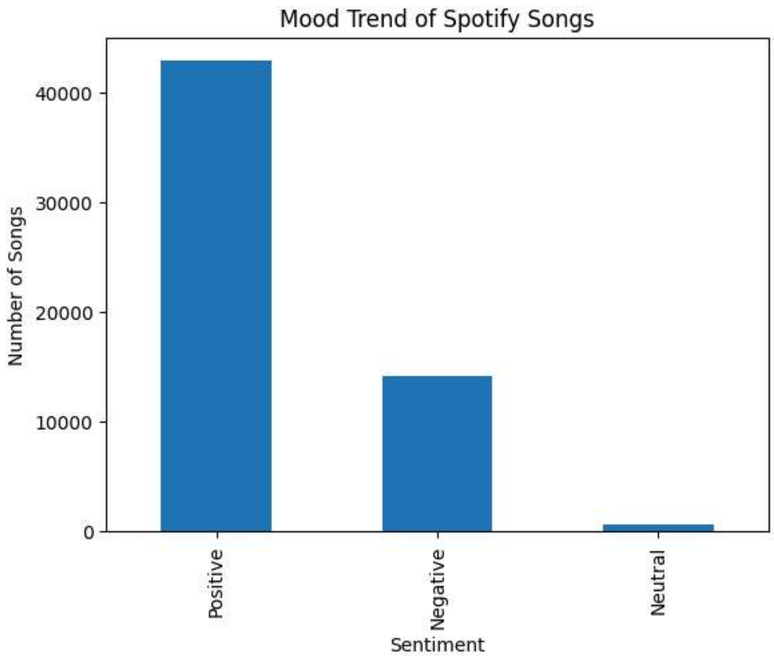
	song	artist	Sentiment
0	Ahe's My Kind Of Girl	ABBA	Positive
1	Andante, Andante	ABBA	Positive
2	As Good As New	ABBA	Positive
3	Bang	ABBA	Positive
4	Bang-A-Boomerang	ABBA	Positive

```
df['Sentiment'].value_counts()
```

Sentiment	count
Positive	42909
Negative	14110
Neutral	631

dtype: int64

```
import matplotlib.pyplot as plt

# Count sentiments
sentiment_counts = df['Sentiment'].value_counts()

# Plot graph
sentiment_counts.plot(kind='bar')

plt.title('Mood Trend of Spotify Songs')
plt.xlabel('Sentiment')
plt.ylabel('Number of Songs')

plt.show()
```




```
from textblob import TextBlob

# Calculate polarity score
df['Polarity'] = df['text'].apply(
    lambda x: TextBlob(str(x)).sentiment.polarity
)

# Top positive songs
top_positive = df[['song', 'artist', 'Polarity']].sort_values(
    by='Polarity',
    ascending=False
)

top_positive.head(10)
```

	song	artist	Polarity
24166	Three Coins In The Fountain	Andy Williams	1.0
19931	Legalise It	UB40	1.0
46000	The Perfect Drug (Meat Beat Manifesto)	Nine Inch Nails	1.0
36155	Eden	Hooverphonic	1.0
15511	That Wonderful Someone	Patsy Cline	1.0
27463	All Together Now	Children	1.0
43363	Goin' Up Yonder - Angel Burgess	Mc Hammer	1.0
6729	Brothers Of The Highway	George Strait	1.0
42092	Real Good Time Together	Lou Reed	1.0
30393	Baby Doll	Doris Day	1.0

```
# Top negative songs
top_negative = df[['song', 'artist', 'Polarity']].sort_values(
    by='Polarity'
)

top_negative.head(10)
```

	song	artist	Polarity
10795	I Wanna Be Sedated	Kirsty Maccoll	-1.000000
21655	The King Will Come	Wishbone Ash	-1.000000
44171	Scream!	Misfits	-1.000000
48259	F-Games	Phineas And Ferb	-1.000000
39456	Watching You Without Me	Kate Bush	-1.000000
50688	It Happened Today	Rem	-1.000000
14715	I Wanna Be Sedated	Offspring	-1.000000
54992	The Evil One	Venom	-0.866667
31826	Kind Hearted Woman	Eric Clapton	-0.800000
44181	The Haunting	Misfits	-0.800000

```
!pip install wordcloud
```

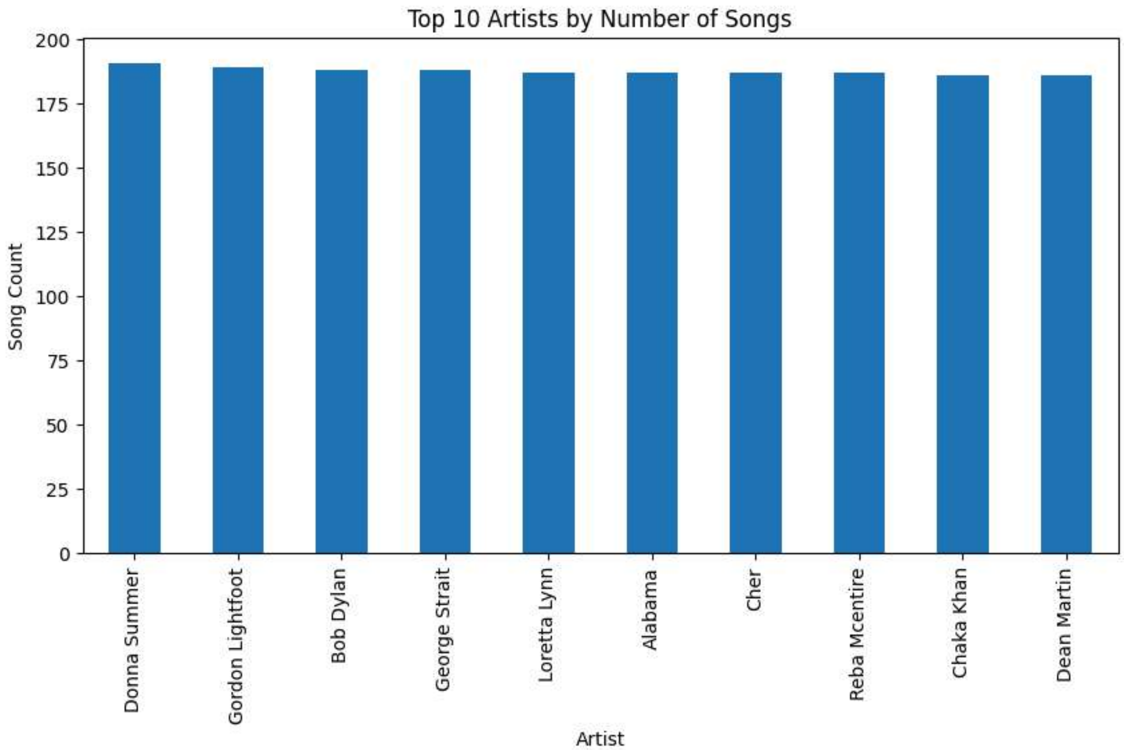
```
Requirement already satisfied: wordcloud in /usr/local/lib/python3.12/dist-packages (1.9.6)
Requirement already satisfied: numpy>=1.19.3 in /usr/local/lib/python3.12/dist-packages (from wordcloud) (2.0.2)
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packages (from wordcloud) (11.3.0)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (from wordcloud) (3.10.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (4.62.1
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (26.1)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib->wordcloud) (2.9
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib->wordc
```

```
from wordcloud import WordCloud
import matplotlib.pyplot as plt

# Combine all lyrics
all_words = ' '.join(df['text'].astype(str))

# Create word cloud
wordcloud = WordCloud(
    width=800,
    height=400,
    background_color='white'
).generate(all_words)

# Display word cloud
plt.figure(figsize=(15,8))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.title('Most Common Words in Spotify Song Lyrics')

plt.show()
```

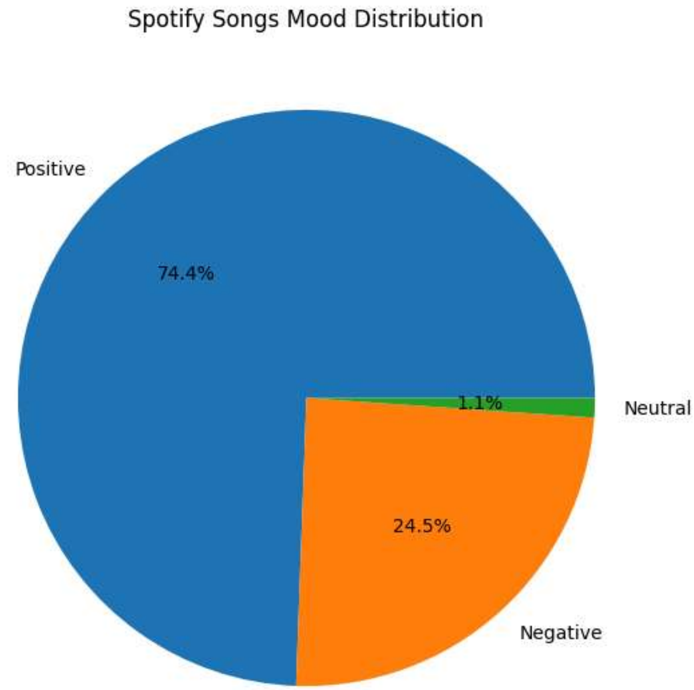

```
# Count sentiments by artist
artist_sentiment = df.groupby(['artist', 'Sentiment']).size().unstack()

artist_sentiment.head()
```

Sentiment artist	Negative	Neutral	Positive
'n Sync	17.0	1.0	75.0
ABBA	25.0	1.0	87.0
Ace Of Base	10.0	NaN	64.0
Adam Sandler	22.0	NaN	48.0
Adele	14.0	NaN	40.0

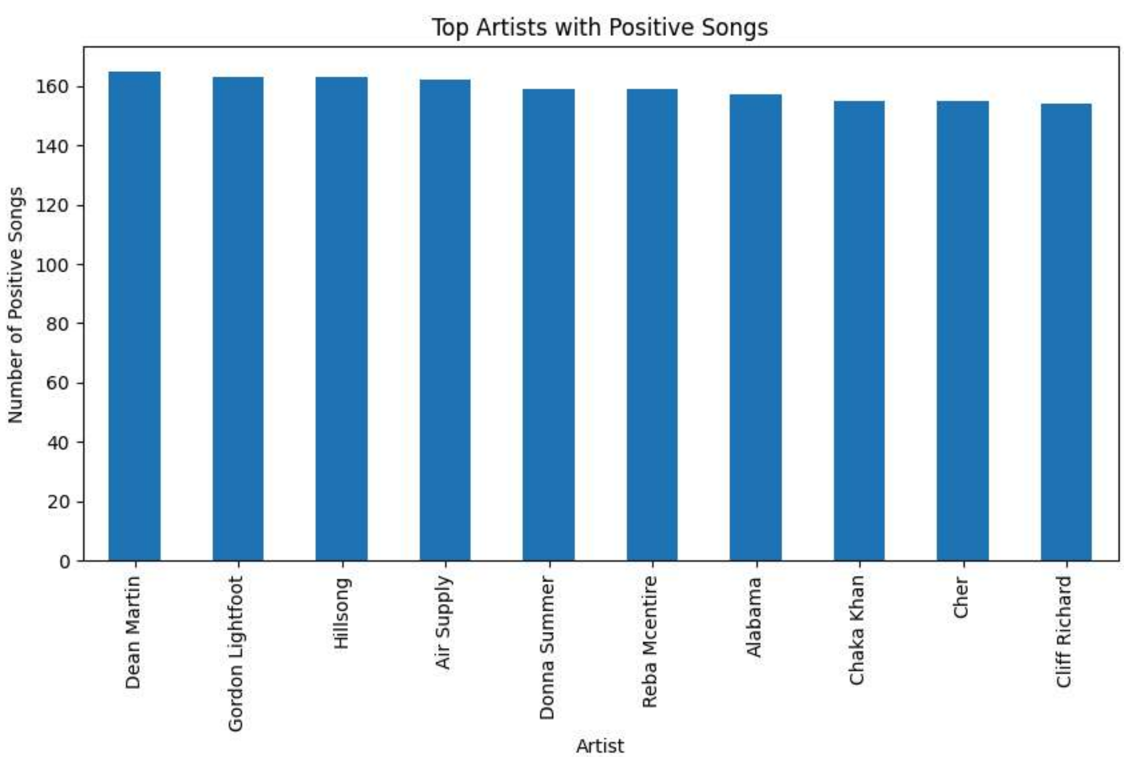
```
# Top artists with most songs
top_artists = df['artist'].value_counts().head(10)

top_artists.plot(kind='bar', figsize=(10,5))

import matplotlib.pyplot as plt

plt.title('Top 10 Artists by Number of Songs')
plt.xlabel('Artist')
plt.ylabel('Song Count')

plt.show()
```




```python
import matplotlib.pyplot as plt

# Sentiment counts
sentiment_counts = df['Sentiment'].value_counts()

# Pie chart
plt.figure(figsize=(7,7))

plt.pie(
    sentiment_counts,
    labels=sentiment_counts.index,
    autopct='%1.1f%%'
)

plt.title('Spotify Songs Mood Distribution')

plt.show()
```




```
# Top 10 positive artists
positive_artists = df[df['Sentiment'] == 'Positive']['artist'].value_counts().head(10)

plt.figure(figsize=(10,5))

positive_artists.plot(kind='bar')

plt.title('Top Artists with Positive Songs')
plt.xlabel('Artist')
plt.ylabel('Number of Positive Songs')

plt.show()
```



```
!pip install spotipy
```

```
Collecting spotipy
  Downloading spotipy-2.26.0-py3-none-any.whl.metadata (5.1 kB)
Collecting redis>=3.5.3 (from spotipy)
  Downloading redis-7.4.0-py3-none-any.whl.metadata (12 kB)
Requirement already satisfied: requests>=2.25.0 in /usr/local/lib/python3.12/dist-packages (from spotipy) (2.32.4)
Requirement already satisfied: urllib3>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from spotipy) (2.5.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.25.0->spoti
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.25.0->spotipy) (3.13)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.25.0->spotipy) (2
Downloading spotipy-2.26.0-py3-none-any.whl (31 kB)
Downloading redis-7.4.0-py3-none-any.whl (409 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 409.8/409.8 kB 9.7 MB/s eta 0:00:00
Installing collected packages: redis, spotipy
Successfully installed redis-7.4.0 spotipy-2.26.0
```

```
import spotipy
from spotipy.oauth2 import SpotifyClientCredentials
```

```
!pip install nrclex
```

```
Collecting nrclex
  Downloading nrclex-4.1.0-py3-none-any.whl.metadata (2.7 kB)
Requirement already satisfied: textblob>=0.17 in /usr/local/lib/python3.12/dist-packages (from nrclex) (0.19.0)
Requirement already satisfied: nltk>=3.9 in /usr/local/lib/python3.12/dist-packages (from textblob>=0.17->nrclex) (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk>=3.9->textblob>=0.17->nrclex) (8.3.3)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk>=3.9->textblob>=0.17->nrclex) (1.5.3
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk>=3.9->textblob>=0.17->nrcle
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk>=3.9->textblob>=0.17->nrclex) (4.67.3)
Downloading nrclex-4.1.0-py3-none-any.whl (44 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 44.5/44.5 kB 1.3 MB/s eta 0:00:00
Installing collected packages: nrclex
Successfully installed nrclex-4.1.0
```

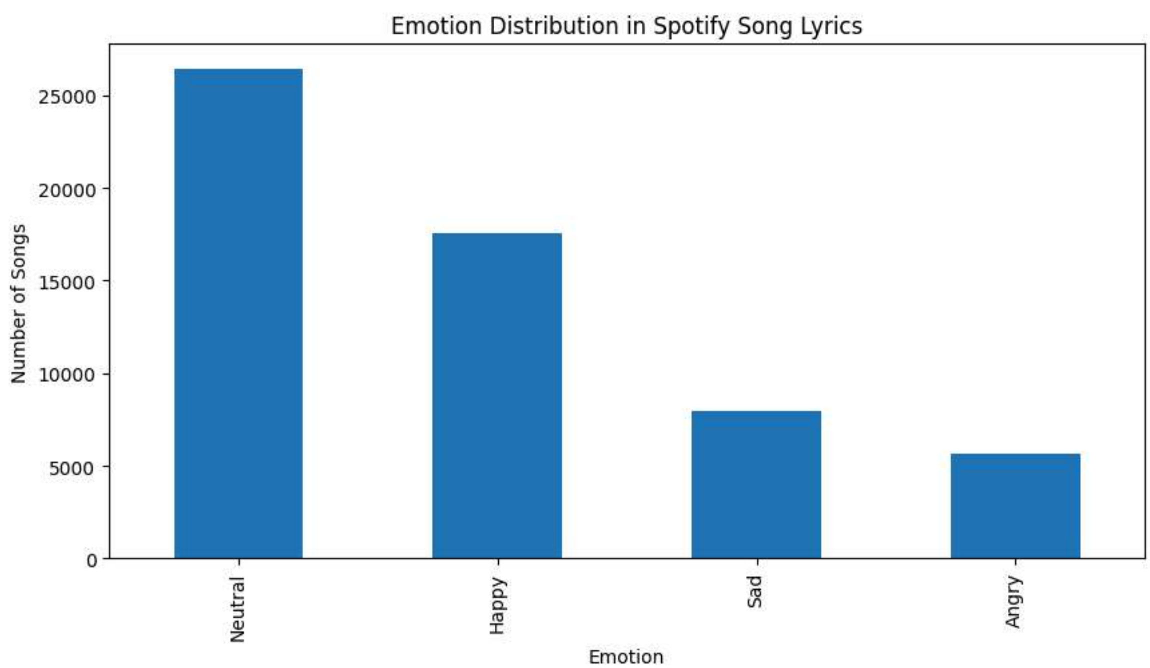
```
import nltk

nltk.download('punkt')
nltk.download('stopwords')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
True
```

```
# Simple emotion detection using keywords

def detect_emotion(text):
    text = str(text).lower()

    happy_words = ['love', 'happy', 'joy', 'smile', 'fun']
    sad_words = ['cry', 'sad', 'pain', 'alone', 'hurt']
    angry_words = ['hate', 'anger', 'mad', 'fight', 'rage']

    happy_score = sum(word in text for word in happy_words)
    sad_score = sum(word in text for word in sad_words)
    angry_score = sum(word in text for word in angry_words)

    if happy_score > sad_score and happy_score > angry_score:
        return 'Happy'
    elif sad_score > happy_score and sad_score > angry_score:
        return 'Sad'
    elif angry_score > happy_score and angry_score > sad_score:
        return 'Angry'
    else:
        return 'Neutral'

# Create Emotion column
df['Emotion'] = df['text'].apply(detect_emotion)

# Check results
df[['song', 'artist', 'Emotion']].head()
```

	song	artist	Emotion
0	Ahe's My Kind Of Girl	ABBA	Happy
1	Andante, Andante	ABBA	Neutral
2	As Good As New	ABBA	Happy
3	Bang	ABBA	Happy
4	Bang-A-Boomerang	ABBA	Happy

```
import matplotlib.pyplot as plt

# Count emotions
emotion_counts = df['Emotion'].value_counts()

# Plot graph
plt.figure(figsize=(10,5))

emotion_counts.plot(kind='bar')

plt.title('Emotion Distribution in Spotify Song Lyrics')
plt.xlabel('Emotion')
plt.ylabel('Number of Songs')

plt.show()
```




```
# Top artists by number of songs
top_artists = df['artist'].value_counts().head(5).index

# Filter dataset
artist_df = df[df['artist'].isin(top_artists)]

# Emotion count by artist
emotion_artist = artist_df.groupby(['artist', 'Emotion']).size().unstack(fill_value=0)

emotion_artist
```

Emotion Angry Happy Neutral Sad

```
emotion_artist.plot(kind='bar', figsize=(12,6))

import matplotlib.pyplot as plt

plt.title('Emotion Comparison Between Top Artists')
plt.xlabel('Artist')
plt.ylabel('Number of Songs')

plt.show()
```